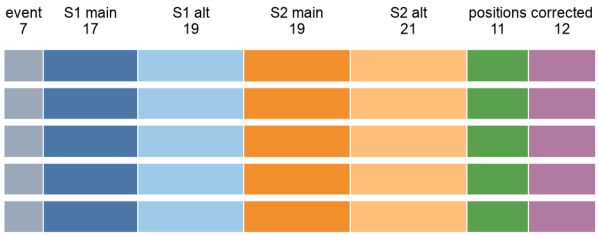
Proposal: a structured event data model in amstrax

For discussion with the XAMS group. Version 2, 01-11-2026 (v1: 31-10-2026). No code yet. Sizes measured on PROD3 run 7790 (Cs, 97 V/cm; 173 651 events, 779 905 peaks).

In short. Replace the flat main / alt columns by **four linked tables**: **events** (the event as a whole: counts, total energy, neighbours), **sites** (one row per S2 interaction site: position and corrected S2), **event_peaks** (every peak inside the event, small ones included) and **peak_context** (every peak of the run, linked to the last large S2 before it). Each quantity is stored once; no main / alt names. The event level gets ~30–40 % smaller. Introduced as optional data types first; the switch is a separate decision.

1 Now and proposed

now: event_info, one row per event, 104 columns



- the same peak quantities 4× (main / alt S1, main / alt S2)
- room for 2 S1s and 2 S2s: a 3rd S2 is lost
- event_info is a stored merge of event_basics, positions and corrected areas: stored twice
- small peaks and the tail after the event: only in peak_basics, without a link to the event

proposed: four linked tables

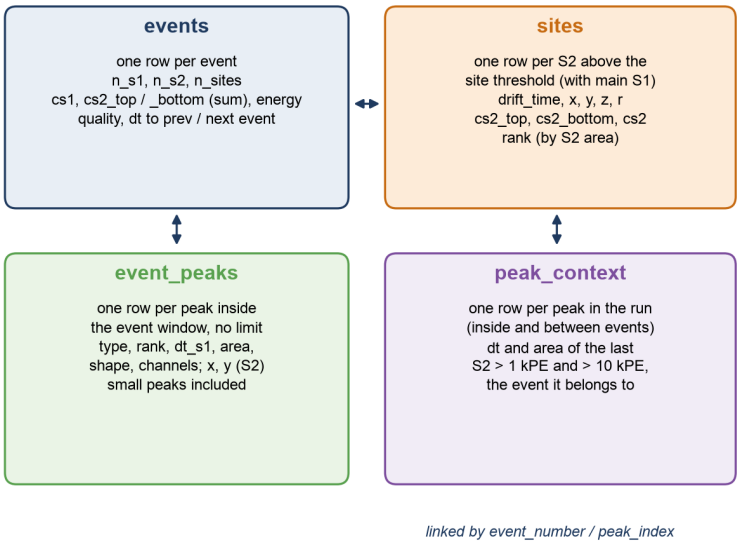


Figure 1. Left: the 104 columns of event_info in blocks. Right: the proposed tables.

2 Why sites, and where the energy lives

A gamma that scatters three times gives one S1 (the scatters are ns apart) and three S2s (different depths). Each S2 is one **site** with its own drift time, x, y, z and corrected S2. The S1 cannot be split over the sites, so cS1 and the energy belong to the **event**: E = W (cS1 / g1 + Σ cS2_top,i / g2), with cS1 corrected for the S1 light collection averaged over the sites (S2-weighted). For a single scatter this is the familiar energy. Small S2-like peaks (delayed electrons, photoionisation) are not sites and are not in the energy: their charge is already in the large S2s.

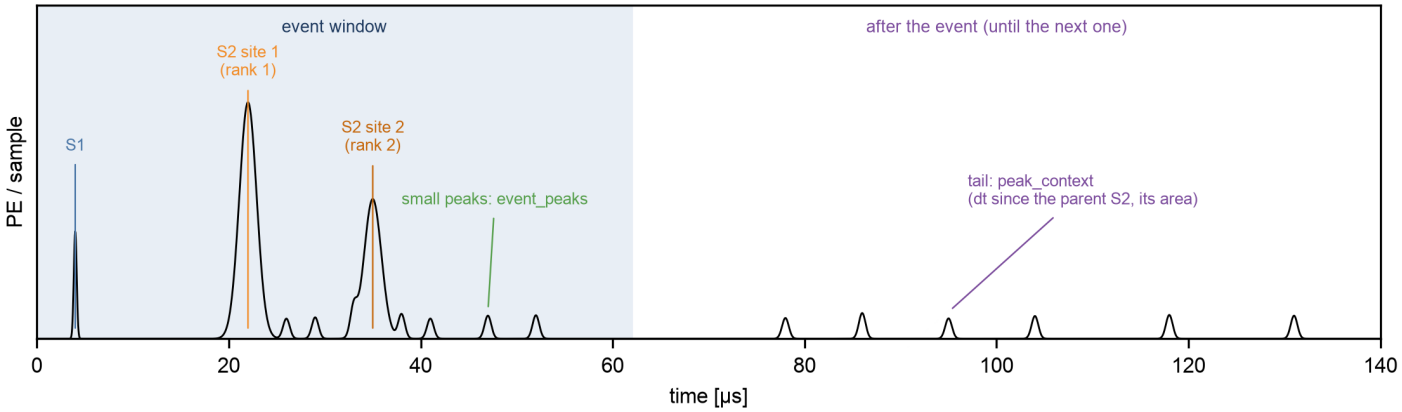


Figure 2. One S1, two sites, small peaks inside the event window (event_peaks) and the tail after it (peak_context).

3 The tables

events (one row per event)	columns
identity	time, endtime, event_number
content	n_peaks, n_s1, n_s2, n_sites
energy	cs1 (S1 corrected with the S2-weighted position of the sites), cs2_top, cs2_bottom, cs2 (sums over sites), energy (from cs2_top)
quality	area_before_main_s2, large_s2_before_main_s2
neighbours	dt_prev_event, dt_next_event (exposure for delayed signals)

No main-S1 / main-S2 columns: those are the rank-1 rows of event_peaks and sites.

sites (one row per S2 above the site threshold)	columns
link	event_number, peak_index, rank (by S2 area; 1 = main)

sites (one row per S2 above the site threshold)	columns
position	drift_time (from the main S1), x, y, z, r
corrected	cs2_top, cs2_bottom, cs2 (own position and depth; bottom can saturate for large S2s)
flags	dt to the previous S2, width vs depth (split S2, photoionisation at t_max)
event_peaks (one row per peak in the event)	**columns**
link	event_number, peak_index (row in peaks), type (1 S1, 2 S2, 0 other), rank (by area within type)
time	time, endtime, center_time, dt_s1 (time after the main S1)
size / shape	area, area_fraction_top, range_50p_area, range_90p_area, rise_time
channels	n_channels, n_hits, max_pmt, max_pmt_area, tight_coincidence, n_saturated_channels
position (S2)	x, y
peak_context (one row per peak in the run)	**columns**
link	peak_index; event_number (−1 between events)
parent S2	dt and area of the last S2 > 1 kPE, and of the last S2 > 10 kPE; prev_event (event of that S2)

peak_context links to the last *large S2*, not to the previous event: delayed electrons of a 50 kPE S2 last ms, and a small event in between is not their source. Two thresholds show whether a smaller S2 came in between.

4 How do I …

question	now (event_info)	proposed (ev = events, s = sites, p = event_peaks, pc = peak_context)
main interaction (flat)	`e = event_info`	`amstrax.main_interaction(st, run) (join, not stored)`
2nd largest S2	`e['alt_s2_area']`	`s[s['rank'] == 2] (sites) or p[(p.type == 2) & (p.rank == 2)]`
3rd largest S2	`not available`	`s[s['rank'] == 3] (sites) or p[(p.type == 2) & (p.rank == 3)]`
all S2s > 500 PE	`not available`	`p[(p.type == 2) & (p.area > 500)]`
S2 width of a site	`only main / alt S2`	`s.merge(p[['peak_index', 'range_50p_area']], on='peak_index') (sites + event_peaks)`
single scatters	`approximate (alt_s2 < cut)`	`ev[ev.n_sites == 1]`
energy of a double scatter	`main S2 only`	`ev.energy (S1 + all sites)`
small peaks in the event	`peak_basics + own matching`	`p[p.event_number == 17]`
delayed signals after S2s	`own script (parents, exposure)`	`pc[(pc.dt_prev_s2_10k > 0) & (pc.dt_prev_s2_10k < 2000)]`

The flat main-interaction table that most plots use (s1_area, s2_area, x, y, z, cs1, cs2_top, energy) comes from one amstrax function: a join over the tables, not stored.

5 Size (PROD3 run 7790, blosc as strax stores it)

data	rows	MB	note
raw_records		4 970	90 % of the run; unchanged
peaks / peak_basics / peak_positions	780 k peaks	395 / 39 / 29	unchanged
now, event level	174 k events	**≈ 88**	event_basics 34 + event_positions 8.7 + corrected 4.0 + event_info 41 (stored merge)
events (slim)	174 k	7.5	14 columns, 68 B/row
sites (S2 > 300 PE)	285 k (1.64 / event)	5.6	many 2nd S2s are trains / split S2s: threshold matters
event_peaks	576 k (3.32 / event)	38.2	23 columns, 96 B/row
proposed, event level		**≈ 51**	−40 %
peak_context	780 k (all peaks)	11.5	204 k of them between events; with it ≈ 63 MB (−30 %)

The total per run (≈ 5.5 GB) hardly changes: the event level is ~1.5 % of it. During the optional phase (both formats) the extra is ~50–60 MB per run (~1 %). Estimate built in memory from PROD3 data; x / y / z were filled with random numbers (compress worse than real values, so slightly pessimistic). Runs with lower thresholds or larger S2s have more peaks.

6 Introduction: optional first

step	what
1	Agree on the tables, columns and the open choices (this note).
2	amstrax: new plugins events_v2 / sites / event_peaks / peak_context and the loader main_interaction(). They only *read* existing data types (peak_basics, peak_positions, event_basics), so the hashes of existing data do not change. Not in the auto-processing targets: made only on request.
3	Try: compute them privately from the stored PROD3 peaks (no raw data, no central storage) and rewrite one or two analyses with them.
4	Decision (user). Then: add them to the targets of a new version (PROD4), dashboard event view and handbook, notebooks move with PROD4.
5	A later version drops event_info and the alt_* columns from the targets. PROD0–3 keep their stored event_info.

Breaks after the switch: code that uses alt_* names, the 3 copies of x / y or event_info directly (≈ 150 files in the SR analyses, the dashboard dataview, the event displays). Most notebooks change only their load line to main_interaction().

7 Open choices

question	options (first proposal in bold)
site threshold	**minimum S2 area (a few 100 PE, scales with anode voltage)**, from the 2nd-largest-S2 spectrum; optionally a width-vs-depth check

question	options (first proposal in bold)
fake 2nd sites (split S2, photoionisation at t_max, trains)	**keep as sites, with flags**, or leave out of sites
which S2s pair with which S1	**all sites with the main S1; flag n_s1 > 1 (pile-up)**, or every S1-S2 pair
rank	**by S2 area** (independent of corrections), or by energy
small peaks in event_peaks	**all** (event window bounds the number), or a minimum area
peak_context thresholds	**1 kPE and 10 kPE**, or set per analysis
single electrons	below peak_min_area (10 PE) and the 15 ADC hit threshold: not in any table. Lower them centrally (more peaks), or keep photon-level reprocessing for dedicated studies